The 5 best real-world cases for your infographic



1. AIIMS Delhi, India — 2022

Indian case study

Attack: Ransomware

Patient registration, hospital records and digital services were disrupted. The main server facility resumed service after about two weeks, and officials later reported data recovery from an unaffected backup.



2. NHS, United Kingdom — 2017

Attack: WannaCry ransomware worm

The attack spread across vulnerable Windows computers, disrupting hospital services and cancelling thousands of appointments. The UK government estimated the NHS impact at approximately £92 million.



3. Maersk, international shipping — 2017

Attack: NotPetya malware

Malware spread through company systems, disrupting shipping and port operations. The estimated financial impact was US$300 million. Rebuilding thousands of IT systems took an intensive recovery effort.



4. Saudi Aramco, Saudi Arabia — 2012

Attack: Shamoon destructive malware

Approximately 30,000 workstations were affected. The company cleaned and restored its systems, while its isolated oil production control systems remained operational. Main internal network services were restored in about 12 days.



5. Sony Pictures, USA — 2014

Attack: Destructive malware and data theft

Thousands of computers became unusable, confidential information was stolen, and the company's network was taken offline. Sony estimated approximately US$35 million in costs related to restoring IT and financial systems and investigating the attack.

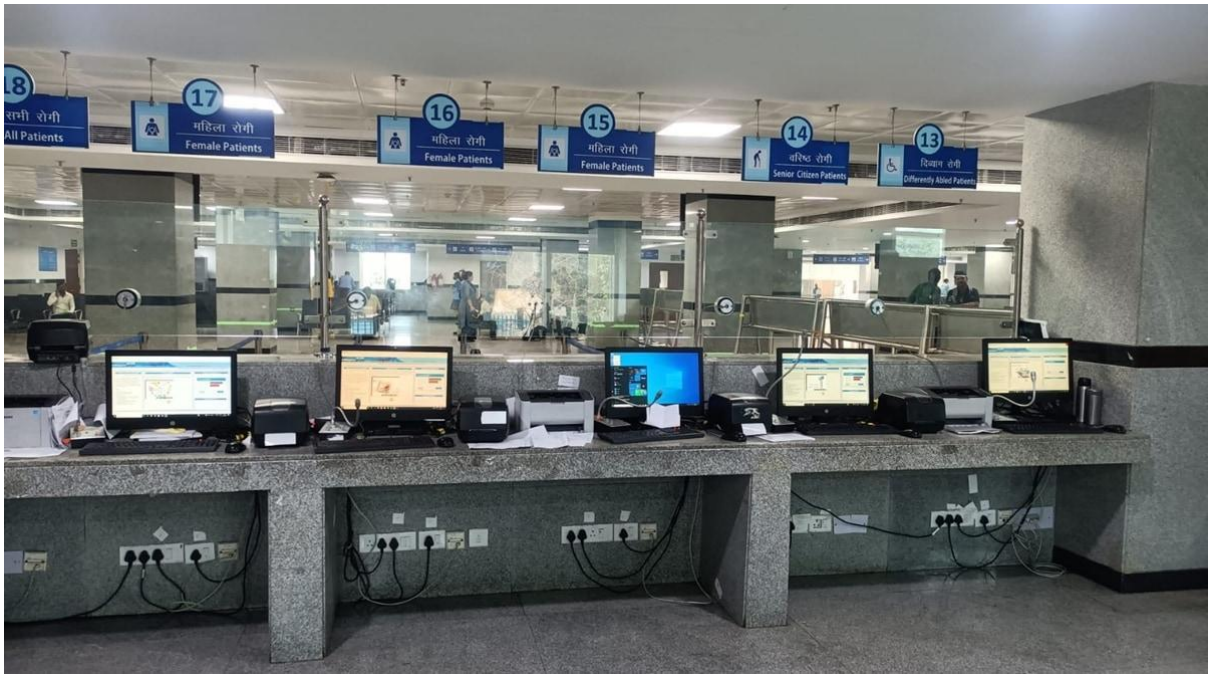
- **Detailed case-study content for your slides**

You can use one slide per case, with the company photo on the left and the facts on the right.

Case 1 — AIIMS Delhi ransomware attack







Date: 23 November 2022 | Location: New Delhi, India

History and how it happened

AIIMS Delhi experienced a ransomware attack that disrupted its e-Hospital system, including patient registration and access to medical records. The specific ransomware family and the precise initial entry method were not publicly confirmed.

Harm and financial loss

- Hospital registration, billing and digital medical services were disrupted.
- Staff had to use manual procedures while systems were unavailable.

- Patient information was at risk.
- Total financial loss: A verified official total was not publicly established.

Recovery time: About two weeks for the main server facility to resume service; further restoration continued afterward.

Recovery actions and cost

- Disconnected and investigated affected systems.
- Sanitised affected servers and computers.
- Retrieved data from an unaffected backup server.
- Worked with government cybersecurity and law-enforcement teams.

Recovery cost: No reliable, publicly confirmed official figure for the full recovery cost was identified.

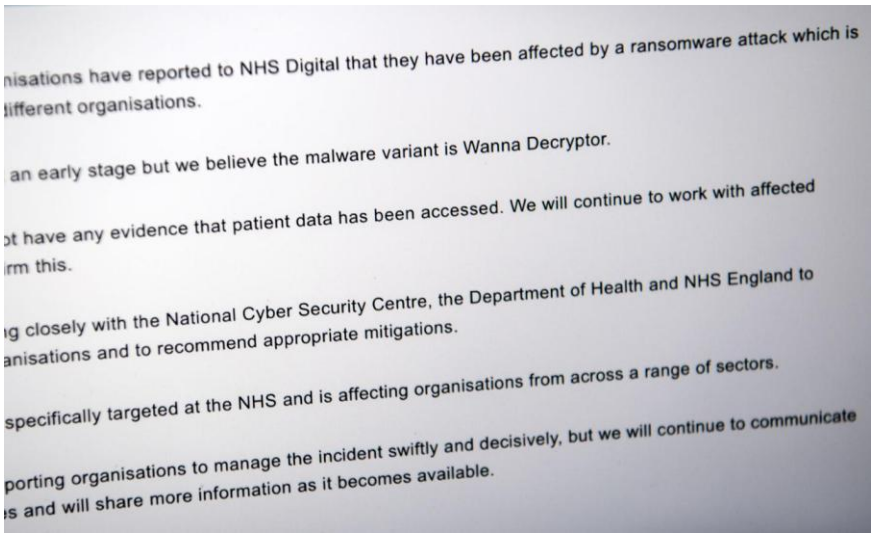
Solutions to prevent recurrence

- Maintain offline and regularly tested backups.
- Use network segmentation and properly configured firewalls.
- Keep operating systems and applications updated.
- Apply strict access controls and monitor suspicious activity.

Case 2 — WannaCry attack on the UK's NHS







Date: 12 May 2017 | Location: United Kingdom and worldwide

History and how it happened

WannaCry was ransomware with worm-like spreading capability. It exploited a Windows vulnerability to spread between vulnerable computers, encrypting files and demanding ransom. Many affected organisations had not applied the available security patch.

Harm and financial loss

- At least 81 of 236 NHS hospital trusts were affected.
- Thousands of appointments and operations were cancelled.
- Some hospitals diverted emergency patients.
- Estimated NHS financial impact: £92 million, according to the UK Department of Health and Social Care.

Recovery time: Emergency disruption was reduced over several days, with broader remediation and follow-up work continuing afterward.

Recovery actions and cost

- Isolated affected systems and restricted network connections.
- Restored services and used telephone or paper-based alternatives.
- Applied security patches and strengthened network protections.
- Used technical support and additional staff effort to restore operations.

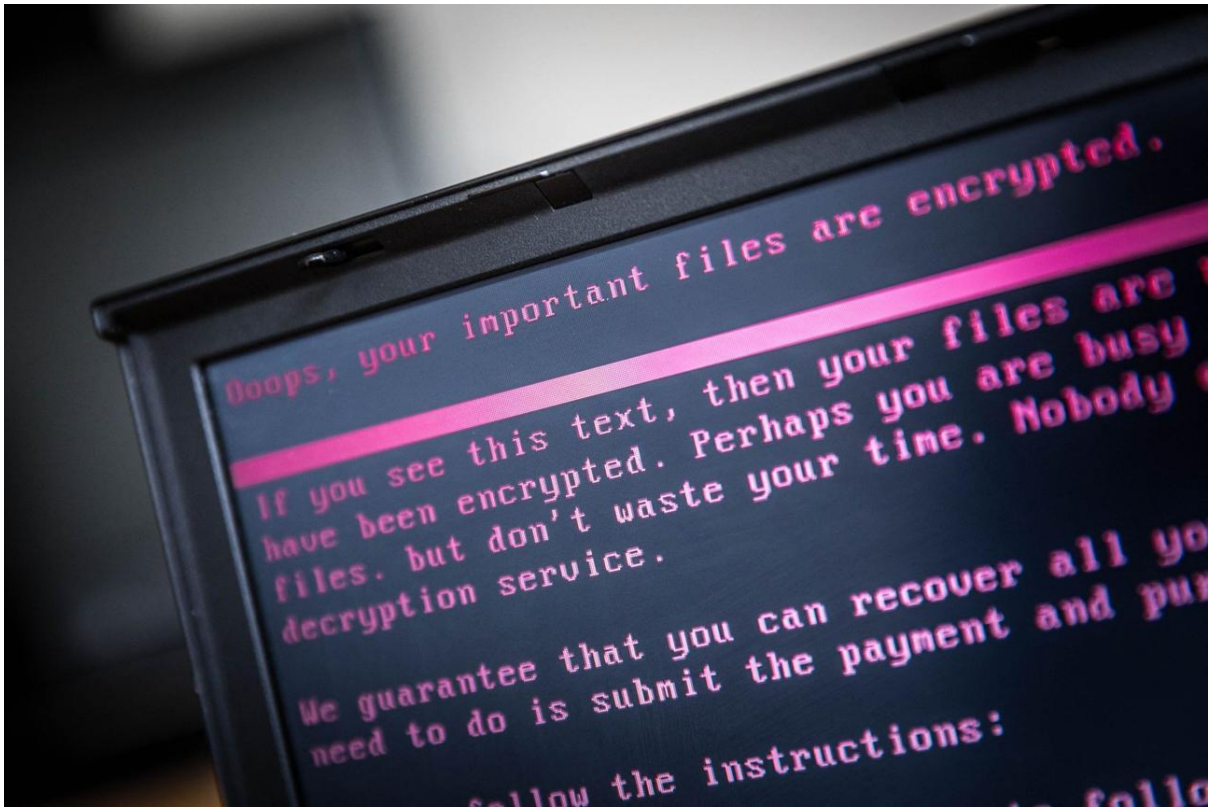
Recovery cost: The NHS did not publish a separately verified recovery-only total. The £92 million is an estimated overall impact, not a confirmed recovery bill.

Solutions to prevent recurrence

- Install security patches promptly.
- Replace unsupported operating systems.
- Maintain tested backups.
- Segment networks and prepare emergency continuity plans.

Case 3 — NotPetya attack on Maersk







Date: 27 June 2017 | Location: Global shipping operations

History and how it happened

NotPetya entered networks through a compromised update mechanism associated with Ukrainian accounting software. It spread through connected systems and caused destructive disruption. Maersk confirmed that its systems were affected by the global Petya cyberattack.

Harm and financial loss

- Shipping and port operations were disrupted.
- Many IT systems had to be rebuilt.
- Staff and customers experienced delays.
- Estimated financial impact: US$300 million.

Recovery time: A major rebuilding effort restored thousands of systems over roughly ten days, although some operational disruption lasted longer.

Recovery actions and cost

- Shut down and contained affected systems.
- Worked with IT partners and cybersecurity specialists.
- Reinstalled approximately 4,000 servers and 45,000 PCs, and rebuilt applications.
- Used manual processes to maintain essential business operations.

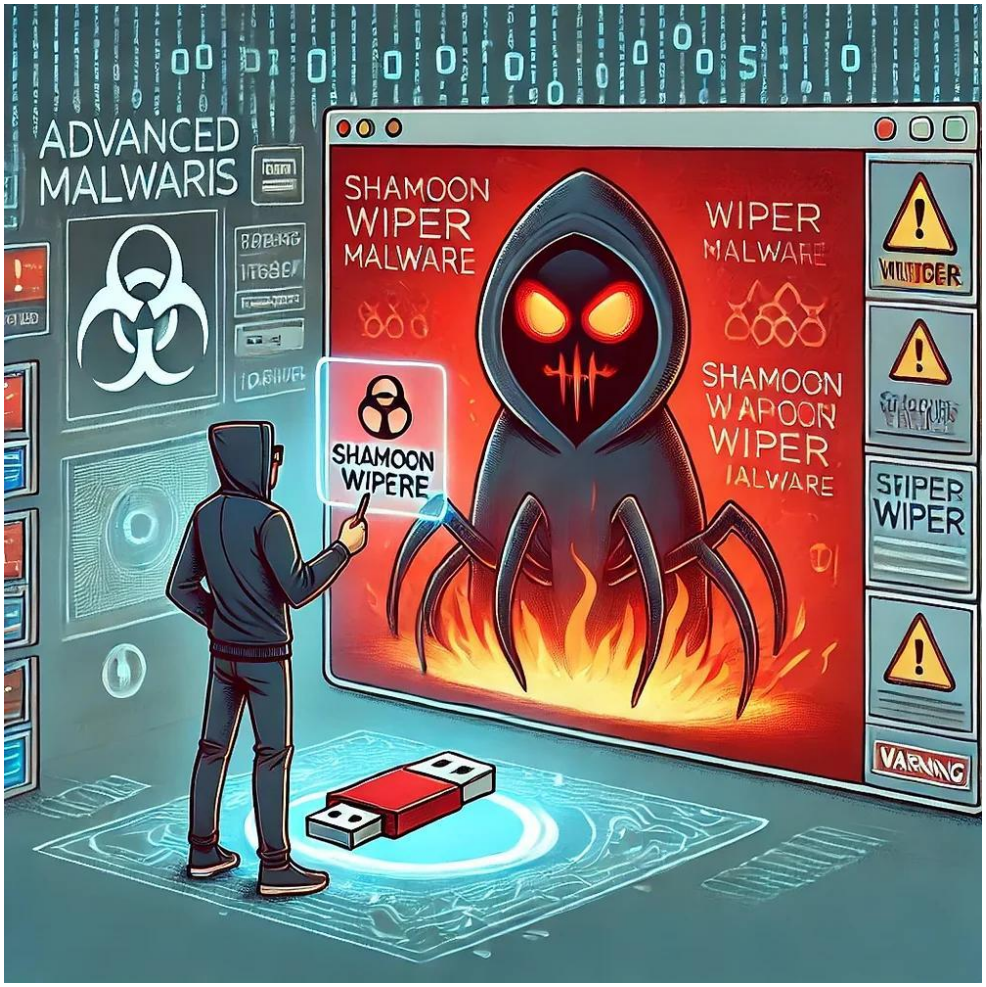
Recovery cost: The widely reported US$300 million represents the estimated overall impact, including business disruption and technology recovery; a separate exact recovery-only cost was not established in the cited sources.

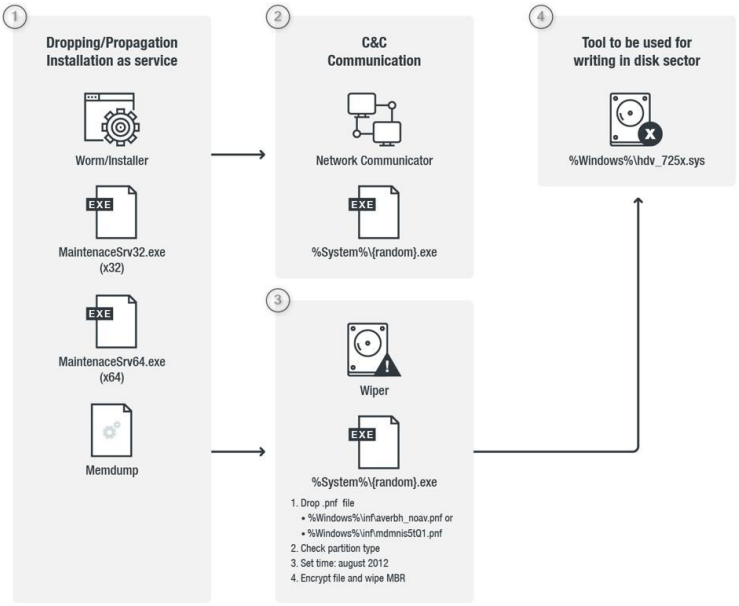
Solutions to prevent recurrence

- Verify software updates and protect software supply chains.
- Restrict administrator privileges.
- Segment networks to contain malware.
- Keep offline backups and test disaster recovery procedures.

Case 4 — Shamoon attack on Saudi Aramco







Date: 15 August 2012 | Location: Saudi Arabia

History and how it happened

Shamoon was destructive malware that overwrote data on infected computers, rendering them unusable. It affected approximately 30,000 Saudi Aramco workstations. The attack was attributed to an organised external group in contemporary investigations, although attribution details have varied.

Harm and financial loss

- Thousands of business computers lost their data.
- Internal office and business services were disrupted.
- Oil production continued because critical production-control systems were isolated from the affected network.

Financial loss: A reliable, publicly confirmed total for the incident was not established in the cited sources.

Recovery time: About 12 days, from the attack on 15 August until the announcement that main internal network services were restored on 27 August 2012.

Recovery actions and cost

- Cleaned infected workstations and restored them to service.
- Restricted remote internet access as a precaution.
- Restored internal network services.
- Kept production-control systems isolated.

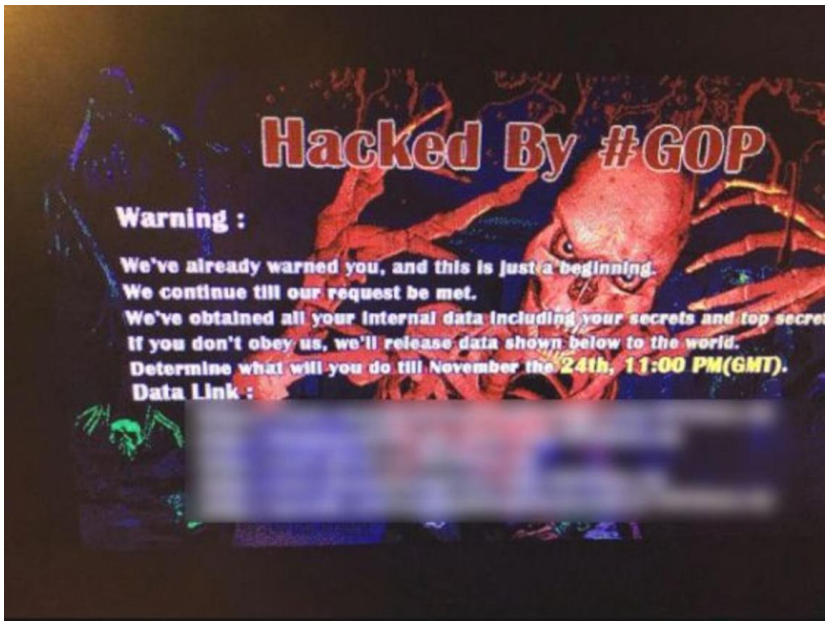
Recovery cost: No verified public figure for the full recovery cost was identified.

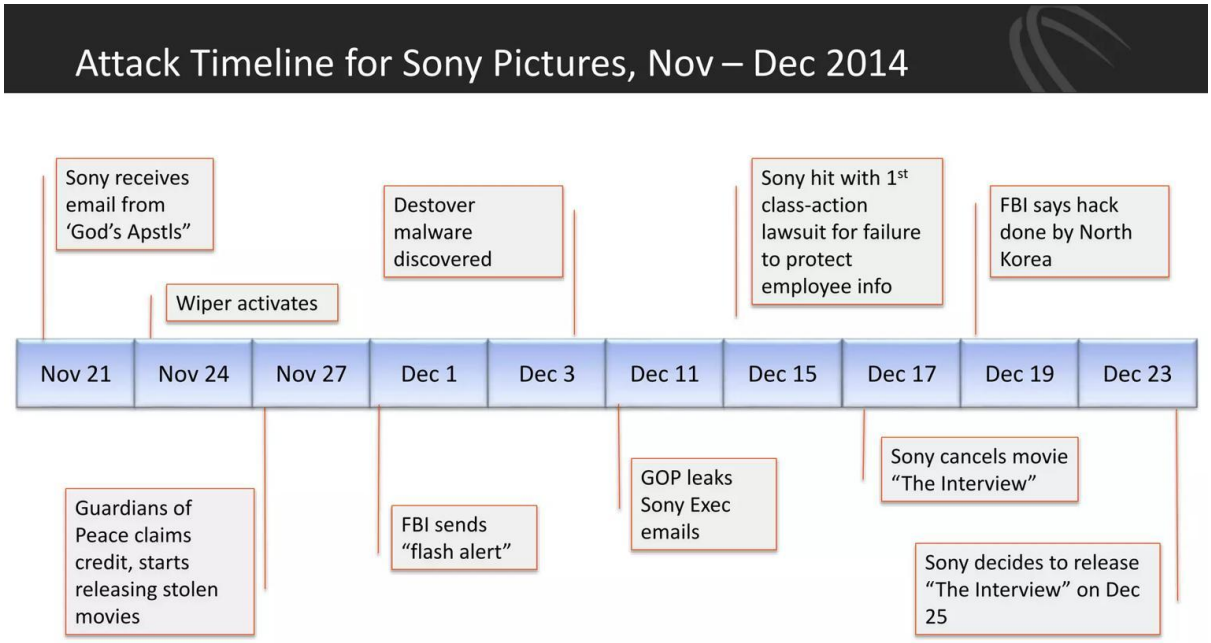
Solutions to prevent recurrence

- Separate critical industrial systems from office networks.
- Maintain offline backups and clean recovery images.
- Apply strict access controls.
- Monitor endpoints and restrict unnecessary remote access.

Case 5 — Sony Pictures cyberattack







Date: November 2014 | Location: United States

History and how it happened

Sony Pictures suffered a destructive cyberattack involving malware and the theft of confidential information. Thousands of computers became inoperable, and the company took its network offline. The FBI publicly attributed the attack to North Korea.

Harm and financial loss

- Confidential employee and business information was stolen and disclosed.
- Company computers and IT systems were damaged.
- Internal communications and business operations were disrupted.
- Estimated IT restoration and investigation costs: US$35 million for the fiscal year ending March 2015.

Recovery time: The disruption continued for weeks, but a single, authoritative date for complete recovery was not established in the cited sources.

Recovery actions and cost

- Took the affected network offline.
- Investigated the intrusion and information theft.
- Rebuilt and restored IT and financial systems.
- Worked with investigators and technical specialists.

Recovery cost: Sony's US$35 million estimate covered restoration and investigation-related costs; it should not be treated as a recovery-only figure.

Solutions to prevent recurrence

- Use endpoint protection and network monitoring.
- Apply least-privilege access.
- Secure sensitive data with appropriate encryption.
- Maintain isolated backups and incident-response plans.
- Train employees to recognise suspicious activity.

Comparison table for your infographic

Case	Main malware/attack	Reported impact	Recovery duration
AIIMS Delhi (India)	Ransomware	Official total loss not confirmed	About 2 weeks for main server resumption
NHS (UK)	WannaCry worm/ransomware	£92 million estimated NHS impact	Days for immediate disruption; longer remediation
Maersk (Global)	NotPetya	US$300 million estimated impact	About 10 days for major system rebuilding
Saudi Aramco	Shamoon wiper malware	30,000 workstations affected	About 12 days for main internal services
Sony Pictures (USA)	Destructive malware and data theft	US$35 million estimated IT/investigation costs	Weeks of disruption; exact full recovery date unclear

5 MAJOR CYBERATTACKS

Real-world incidents • Damage • Recovery • Prevention



1. AIIMS — India

Ransomware



2. NHS — UK

WannaCry worm



3. Maersk — Global

NotPetya



4. Saudi Aramco

Shamoon



5. Sony Pictures — USA

Destructive malware and data theft

THE ATTACK CYCLE

Infection

Damage

Recovery

PREVENTION IS CHEAPER THAN RECOVERY

Patch systems • Back up data • Segment networks • Train employees

Sources to cite in your presentation

Use these credible reports and official statements for your bibliography:

- AIIMS Delhi: The Indian Express — AIIMS server recovery and NDTV — Government update on data restoration.
- NHS WannaCry: UK National Audit Office report.
- Maersk: Maersk interim report on the 2017 cyberattack.
- Saudi Aramco: Saudi Press Agency — restoration announcement.
- Sony Pictures: US Department of Justice — Sony investigation and Computerworld — estimated restoration and investigation costs.